A uniform Burnside–Gautschi bound for interpolated gamma values

Research proof and Lean formalization
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Abstract

We prove Conjecture 37 of Schmidlin's paper on Stirling–Gautschi inequalities. The proposed upper bound holds strictly for every nonnegative base point and, more generally, every nonnegative interpolation parameter. The argument combines a classical three-term Stirling remainder bound with a logarithmic entropy estimate. The remaining rational inequality is certified by a quadratic discriminant whose coefficients become positive after translation. The Lean proof includes the literal gamma-function expression and a proof of the classical remainder bound.

1 Statement and reduction

Following Schmidlin [1], put

$$\widehat{\Pi}(x,\alpha)=\Gamma(x+1)(x+1)^{\alpha},\qquad S_{1/2}(y)=\sqrt{2\pi}\,e^{-1/2}(y+1/2)^{y+1/2}e^{-y},$$

and $\widehat{m}(x,\alpha)=\log(\widehat{\Pi}(x,\alpha)/S_{1/2}(x+\alpha))$. Conjecture 37 asks for the following weak inequality when $x\ge 0$ and $0\le\alpha\le 1$. Corollary 32 of the source proves it for $x\ge 1$.

Theorem 1. *For every $x\ge 0$ and $\alpha\ge 0$,*

$$\widehat{m}(x,\alpha)<\frac{1}{12(x+\alpha)+18-12e^{1/48}}.$$

All powers and logarithms in this formula have positive bases. No assertion of optimality is made for the constant.

Set $z=x+1\ge 1$, $a=\alpha-1/2\ge -1/2$, and $k=12(e^{1/48}-1)$. Define the classical Stirling error and an entropy term by

$$R(z)=\log\Gamma(z)-(z-1/2)\log z+z-\tfrac{1}{2}\log(2\pi), \tag{1}$$

$$E(z,a)=(z+a)\log(1+a/z)-a. \tag{2}$$

Direct algebra gives

$$\widehat{m}(x,\alpha)=R(z)-E(z,a),\qquad 12(x+\alpha)+18-12e^{1/48}=12(z+a)-k.$$

The elementary estimates $1+u\le e^{u}\le(1-u)^{-1}$, $u=1/48$, give $1/4\le k<1/2$. In particular, the denominator is positive.

Lemma 2. *If $z>0$ and $z+a>0$, then $E(z,a)\ge 0$. If also $a\ge 0$, then*

$$E(z,a)\ge\frac{a^2}{2z+a}.$$

Proof. Use $\log t\ge 1-t^{-1}$ for the first assertion. For the second, use $\log(1+t)\ge 2t/(2+t)$ with $t=a/z$ and multiply by $z+a$. □

2 An exact positive-polynomial certificate

The classical Stirling remainder estimate [2, §5.11(ii)] gives

$$R(z) \le U(z) := \frac{1}{12z} - \frac{1}{360z^3} + \frac{1}{1260z^5}, \qquad z > 0. \tag{3}$$

For $z \ge 1$, $U(z) \le 1/(12z)$. If $-1/2 \le a < 0$, Lemma 2 therefore gives immediately

$$R(z) - E(z,a) \le \frac{1}{12z} < \frac{1}{12(z+a)-k}.$$

It remains to handle $a \ge 0$.

Define

$$\begin{aligned} A(z) &= 120960z^5, \\ B(z) &= 120960z^6 - 2520z^5 - 10080z^4 + 336z^2 - 96, \\ C(z) &= -20160z^5 + 210z^4 + 1008z^3 - 7z^2 - 288z + 2, \\ F(z) &= 420z^5 + 672z^4 - 14z^3 - 192z^2 + 4z. \end{aligned}$$

Clearing positive denominators yields the identity

$$\frac{1}{12(z+a)-1/4} - U(z) + \frac{a^2}{2z+a} = \frac{A(z)a^3 + B(z)a^2 + C(z)a + F(z)}{2520z^5(a+2z)(48a+48z-1)}. \tag{4}$$

Both $B(z)$ and $\Delta(z) := 4B(z)F(z) - C(z)^2$ are strictly positive for $z \ge 1$. Indeed, writing $z = 1 + t$, $t \ge 0$, their coefficients in descending order are respectively

$$120960,\ 723240,\ 1791720,\ 2353680,\ 1729056,\ 673512,\ 108600$$

and

$$\begin{gathered} 203212800,\ 2149822080,\ 10299502080,\ 29404384380, \\ 55375719840,\ 71883106764,\ 65187512040,\ 40869159443, \\ 17049349100,\ 4345739674,\ 553547260,\ 16630775. \end{gathered}$$

Thus

$$4B(z)\big(B(z)a^2 + C(z)a + F(z)\big) = (2B(z)a + C(z))^2 + \Delta(z) > 0.$$

The cubic term $A(z)a^3$ is nonnegative. Formula (4) and Lemma 2 now imply

$$R(z) - E(z,a) \le U(z) - \frac{a^2}{2z+a} < \frac{1}{12(z+a)-1/4} \le \frac{1}{12(z+a)-k},$$

which proves Theorem 1.

The polynomial identities above are checked in Lean by ring normalization. Their coefficient positivity, the quadratic identity, and every denominator sign are also proved in Lean. The integer lists in this paper are not trusted inputs to an external numerical computation.

3 The classical remainder bound in the formal proof

For completeness, the certificate proves (3) from elementary logarithmic bounds and the pinned library's integer Stirling theorem. The gamma recurrence gives

$$R(z) - R(z+1) = h(z) := (z+1/2)\log(1+1/z) - 1.$$

For $v = (2z+1)^{-1} \in (0,1)$, the atanh-series remainder estimate gives

$$h(z) \le H(z) := \frac{v^2}{3} + \frac{v^4}{5} + \frac{v^6}{7} + \frac{v^8}{1-v^2}.$$

A second exact identity is

$$U(z) - U(z+1) - H(z) = \frac{112z^8 + 448z^7 + 1295z^6 + 2317z^5 + 2200z^4 + 1061z^3 + 253z^2 + 34z + 2}{2520z^5(z+1)^5(2z+1)^6} > 0.$$

Consequently $R(z) - U(z)$ increases under $z \mapsto z+1$. Its limit is zero, so $R(z) \le U(z)$.

Here the real-variable limit is itself proved, rather than assumed as a certificate hypothesis. Log-gamma convexity gives, for integers $n \ge 2$ and $0 \le r \le 1$,

$$\log\Gamma(n) + r\log(n-1) \le \log\Gamma(n+r) \le \log\Gamma(n) + r\log n.$$

Using $u/(1+u) \le \log(1+u) \le u$ yields

$$R(n) - \frac{2}{n-1} \le R(n+r) \le R(n) + \frac{1}{n}.$$

At integers, $R(n)$ is the logarithm of Mathlib's Stirling sequence minus $\frac{1}{2}\log\pi$, and therefore tends to zero. Taking $n = \lfloor z \rfloor$ proves the required limit for real $z \to \infty$.

4 Formal scope and prior work

The Lean endpoint `Schmidlin.source_conjecture_372858` is the complete source conjecture. The stronger endpoint `source_strict_bound` quantifies over every $x \ge 0$ and $\alpha \ge 0$. The definitions use the actual gamma function, real powers, and the literal logarithm of the interpolated-value/Burnside ratio. The logarithmic expansion and all analytic and rational bridges are proved. No classical Stirling or entropy fact is claimed as a new discovery.

The source's established $x \ge 1$ range is credited to Schmidlin. A bounded preliminary search found no existing resolution of the remaining conjecture; it does not certify worldwide priority.

References

[1] M. Schmidlin, *On Gautschi & Stirling identities, asymptotics and inequalities for the Pi (or Gamma) function*, arXiv:2601.21906v1 (2026), Conjecture 37, pp. 28–29. https://arxiv.org/abs/2601.21906v1.

[2] NIST Digital Library of Mathematical Functions, §5.11(i)–(ii), Stirling's series and remainder bounds. https://dlmf.nist.gov/5.11.

[3] The Mathlib Community, *The Lean mathematical library*, pinned revision `d13f23b723b8a846827a245b89c10fc7d3f11612`; `Stirling.lean`, `Gamma/BohrMollerup.lean`.